Bridging Neuroscience and Artificial Intelligence

Student Name: Alfonso Miranda
Course: ITAI-4374
Laboratory Exercise: Bridging Neuroscience and Artificial Intelligence
Submission Date: February 6, 2026

Table of Contents

---

1. Neural Network Comparison Analysis

A biological neuron consists of dendrites, a cell body (soma), an axon, and synapses. Dendrites receive input signals from other neurons, while the soma integrates these signals. If the combined input exceeds a certain threshold, the neuron generates an electrical impulse that travels down the axon and is transmitted to other neurons through synapses. This biological process inspired the design of artificial neural networks (ANNs).

In ANNs, an artificial neuron receives multiple inputs, each multiplied by a weight. These weighted inputs are summed and passed through an activation function to produce an output. While this structure is inspired by biology, artificial neurons are highly simplified. Biological neurons rely on complex electrical and chemical signaling and can perform nonlinear processing within dendrites. Artificial neurons, by contrast, are mathematical abstractions designed for scalability and computational efficiency. Biological systems excel at flexible and adaptive learning, while ANNs perform best in fast, large-scale computations.

Comparative Analysis Table

Biological Neural Network	Artificial Neural Network (ANN)	AI Examples

Dendrites receive signals from many neurons	Inputs/features into a node	Input layers in CNNs and RNNs
Soma integrates signals and triggers output	Weighted sum and activation function	ReLU, sigmoid activations
Axon transmits signals to other neurons	Output passed to next layer	Feedforward networks
Synapses change strength over time	Weights updated during training	Backpropagation
Complex nonlinear biological processing	Simplified mathematical operations	CNNs, RNNs, Transformers

---

2. Brain Architecture and AI Design

The human brain is organized hierarchically, with lower regions handling basic sensory and motor functions and higher regions managing complex cognition. For example, the visual cortex processes visual information in layers, moving from simple edge detection to complex object recognition. The temporal lobe integrates visual input with memory, while the amygdala, part of the limbic system, processes emotions and influences behavior and decision making.

Modern AI architectures reflect these principles. Convolutional Neural Networks (CNNs) mimic the layered processing of the visual cortex by extracting increasingly complex features. Long Short-Term Memory (LSTM) networks resemble hippocampal memory functions by maintaining and updating information over time. Attention mechanisms in transformer models parallel biological attention systems by prioritizing relevant information while ignoring less important inputs.

---

3. Integration Analysis and Future Implications

Understanding biological neural networks can improve AI design by inspiring more efficient and adaptable systems. The brain uses sparse connectivity, parallel processing, and local learning rules, which motivate emerging architectures such as spiking neural networks and neuromorphic computing. These approaches aim to reduce energy consumption and improve temporal learning.

Biological systems are remarkably energy-efficient, operating on roughly 20 watts, while AI systems often require significant computational resources. Humans also learn from very few examples and generalize effectively, whereas AI systems are typically data-intensive and task-specific. However, AI systems excel at speed, scalability, and precision for narrowly defined tasks.

Future AI research may increasingly draw on neuroscience to overcome current limitations. Insights into emotion processing, memory consolidation, attention, and neural plasticity could improve AI decision making, risk assessment, and adaptive learning. Hybrid systems that integrate biological principles with artificial computation—such as neuromorphic chips or brain-computer interfaces—represent promising directions for bridging biological and artificial intelligence.

Note on Visualizations: Any diagrams or images used in this report should be clearly labeled, high resolution, properly referenced if sourced from Wolfram Alpha, and accompanied by explanations linking the visual content to artificial intelligence concepts.